INCH-POUND

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SUPERSEDING
FF-W-84A
17 April 1967

DETAIL SPECIFICATION SHEET

Washers, Lock (Spring)

Inactive for new design after 15 December 2000
and is no longer used, except for replacement purposes.

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

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1. SCOPE

1.1. Scope. This specification covers a helical spring type washer for use with nuts, bolts, screws, and studs to prevent them from loosening.

1.2. Classification. Washers covered by this specification shall be of the following classes and styles, as specified (see 6.2).

Class A – Carbon steel
- Style 1 – Light
- Style 2 – Regular
- Style 3 – Heavy
- Style 4 – Extra duty
- Style 5 – Hi-Collar

Class B – Corrosion-resistant steel (magnetic)
- Style 1 – Light
- Style 2 – Regular
- Style 3 – Heavy

Class C – Corrosion-resistant steel (non-magnetic).
- Style 1 – Light
- Style 2 – Regular
- Style 3 – Hi-Collar

Class D – Aluminum-zinc alloy (7075).
- Style 2 – Regular

Class E – Phosphor bronze or tin brass.
- Style 1 – Light
- Style 2 – Regular

Class G – Nickel-copper-aluminum alloy (K-monel)
- Style 1 – Light
- Style 2 – Regular

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government Documents.

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DEPARTMENT OF DEFENSE STANDARDS

QQ-N-286 – Nickel-Copper-Aluminum Alloy, Wrought (K-Monel)

MIL-PRF-8625 – Anodic Coatings, For Aluminum and Aluminum Alloys

ASTM B545 – Standard Specification for Electrodeposited Coatings of Tin

MIL-DTL-16232 – Phosphate Coatings, Heavy, Manganese or Zinc Based

ASTM A342/A342M – Standard Test Methods for Permeability of Weakly Magnetic Materials

MIL-STD-1916 – DOD Preferred Methods for Acceptance of Product

AIA/NAS NASM35338 – Washer, Lock, Split, Helical, Regular (Medium) Series

MS35340 – Washer, Lock, Split, Helical, Extra Heavy Series

MS51848 – Washer, Lock-Helical Spring, Hi-Collar

(Copies of this documents are available online at https://quicksearch.dla.mil.)

2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ).

ASQ H1331 – Zinc-Nickel Electroplating for Fasteners

(Copies of this document are available for purchase online from www.asq.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE AMS-C-81562 – Coatings, Cadmium, Tin-Cadmium, and Zinc (Mechanically Deposited)

SAE AMS-QQ-A-225/9 – Aluminum Alloy 7075, Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finished

SAE AMS-QQ-P-35 – Passivation Treatments for Corrosion-Resistant Steel

SAE AMS-QQ-P-416 – (R) Plating, Cadmium (Electrodeposited)

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(Copies of this document are available for purchase online from www.sae.org.)

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME) STANDARD.

ASME B18.21.1-1972 – Lock Washers

(Copies of this document are available for purchase online from www.asme.org.)

ASTM INTERNATIONAL (ASTM)

ASTM A276 – Bar and Shape, Stainless and Heat Resisting Steel

ASTM A581/A581M – Standard Specification for Free-Machining Stainless Steel Wire and Wire Rods

ASTM B633 – Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

ASTM A29/A29M – Steel Bars, Carbon and Alloy, Hot-Wrought and Cold-Finished, General Requirements for

ASTM B139/B139M – Standard Specification for Phosphor Bronze Rod, Bar, and Shapes

ASTM A153/153M – Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM B591 – Standard Specification for Copper-Zinc-Tin and Copper-Zinc-Tin-Iron-Nickel Alloys Plate, Sheet, Strip, and Rolled Bar

(Copies of these documents are available for purchase online from www.astm.org.)

2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3. REQUIREMENTS

3.1. Material.

3.1.1. Class A. Unless otherwise specified (see 6.2), class A spring lockwashers made of carbon steel shall be compositions 1060 to 1080 (UNS G10600) to (UNS G10800) in conformance with ASTM A29/A29M. Class A washers shall be properly quenched from above the critical temperature and tempered to a hardness of 38 to 46 HRC. Depending upon the cross section, there may be used the Rockwell C, D, or A scale, the superficial Rockwell N scale or the Vickers

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hardness tester, and the readings converted to Rockwell C. The washers shall meet the decarburization limits of ASME B18.21.1

3.1.2. Class B. Unless otherwise specified (see 6.2), class B spring lockwashers made from corrosion-resisting steel shall be of composition 420 (UNS S42000) in conformance with ASTM A276. After coiling, the class B washer shall be heat treated and quenched in air or oil, as indicated by the proportions of the washer section, and tempered at a temperature not to exceed 800 ˚F to a hardness of 45 to 53 Rockwell C.

3.1.3. Class C. Unless otherwise specified (see 6.2), class C spring lockwashers made of corrosion-resisting steel shall be of the 300 series (UNS S30300) to (UNS S30323) in conformance with ASTM A581/A581M. As the result of cold working during fabrication, the washer shall have a hardness of 35 to 43 Rockwell C. The class C spring lockwashers shall have a magnetic permeability of 2.0 maximum (air=1.0) for a field strength of A=200 Oersteds using a magnetic indicator per ASTM A342/A342M.

3.1.4. Class D. Unless otherwise specified (see 6.2), class D spring lockwashers shall be made of 7075 aluminum alloy in conformance to the requirements of AMS-QQ-A-225/9. The washers shall receive a solution and precipitation heat treatment resulting in a hardness of 75 to 97 rockwell B, or a minimum tensile strength of 77000 PSI. Tensile test specimens of keystone wire heat treated with the washer they represent, shall be furnished when specified in the invitation for bids.

3.1.5. Class E. Unless otherwise specified (see 6.2). class E spring lockwashers shall be made of phosphor bronze in conformance to the requirements of ASTM B139/B139M, composition UNS C51000, or tin brass in conformance with ASTM B591, composition UNS C42500. As the result of cold working during fabrication, class E washers shall have a minimum hardness of 90 Rockwell B.

3.1.6. Class G. Unless otherwise specified (see 6.2), class G spring lockwashers shall be made of nickel-copper-aluminum alloy in conformance to the requirements of QQ-N-286, class A. The washer shall be fabricated from cold-worked material and subsequently age hardened to a hardness of 33 to 40 Rockwell C or a minimum tensile strength of 155,000 PSI.

3.2. Design.

3.2.1. Washer section. The finished washer cross section shall be slightly trapezoidal in shape with the internal and external circumferential edges rounded sufficiently to avoid heat checks. The thickness at the inner periphery shall be greater than the outer periphery by an amount varying from a minimum of 0.0005 to a maximum of 0.001 inch per 1.64 inch of section width. The minimum thickness specified in ASME B18.21.1 represents the nominal mean thickness of the trapezoid. In the formula, the increase in thickness from the outer periphery to the inner periphery shall be from 0.032 W (minimum) to 0.064W (maximum).

3.2.2. Dimensions. When the dimensions are not otherwise specified, styles 1, 2, 3, 4 and 5 spring lockwashers shall conform to those shown in ASME B18.21.1. When specified (see 6.2),

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the dimensions shall conform to the applicable dash numbers of NASM35338, MS35340, and MS51848.

3.2.3. Tolerances. Unless otherwise specified (see 6.2), the tolerance on all dimensions shall be as shown as shown in ASME B18.21.1, or applicable MS standards.

3.2.4. Free height. The washers shall be coiled so that the free height id approximately twice and not less than 1-2/3 times the mean thickness of the washer section. The severed ends shall not overlap or be positioned so that washers will interlink when tested as specified in 4.5.1.

3.3. Protective finishes. Unless otherwise specified (see 6.2), spring lockwashers shall be furnished uncoated. When cadmium, zinc, phosphate, tinned, anodic, or passivation films are specified, they shall conform to the following paragraphs (see 6.2).

3.3.1. Cadmium plating. When specified (see 6.2), cadmium plating shall be in accordance with SAE AMS-QQ-P-416, type II, class 2 or SAE AMS-C-81562, Type II, Class 2 (No substitutes allowed).

3.3.2. Zinc coating. When specified (see 6.2), zinc coting by electrodepositing process shall be in accordance with ASTM B633, Type II, Class Fe/Zn8. Hydrogen embrittlement relief requirement shall be met for washers 38HRC and Higher. Mechanically deposited process in accordance with SAE AMS-C-81562, Type II, class 6 may be used in lieu of electrodepositing process.

3.3.3. Zinc coating (hot-dipped). When specified (see 6.2), zinc coating by hot-dipping process shall be in accordance with ASTM A153/A153M.

3.3.4. Phosphate coating. When specified (see 6.2), phosphate coating shall be in accordance with MIL-DTL-16232, type z, class 2.

3.3.5. Anodic coating. When specified (see 6.2), anodic coating shall be in accordance with MIL-PRF-8625.

3.3.6. Tin coating. When specified (see 6.2), tin coating shall be in accordance with ASTM B545.

3.3.7. Passivation. Passivation treatments shall conform to the requirements of SAE AMS-QQ-P-35.

3.4. Hydrogen embrittlement. Unless otherwise specified, electroplated washers shall be baked for 4 hours at 400 ˚F immediately after plating.

3.5. Cleaners. The cleaners used shall not etch the material and allow subsequent cracking of the washers. Acid cleaners shall not be used. Scale shall be removed by mechanical methods.

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3.6. Workmanship. The flat faces of washers and the inner and outer peripheries shall be smooth and free from knurling, serrations, die marks, deep scratches, etc., although slight feed roll marks are permissible. The washers shall have rounded edges and be free from burrs, rust pit marks, and other defects which may affect their appearance or serviceability. The ends of the gaps shall not present cutting edges at the contact surfaces. The workmanship shall be in accordance with high-grade commercial practice covering this class of work.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2. Lot. A lot shall consist of one type, grade, style, same heat number of material, finish and size on the same contract or order, produced by the same manufacturer under essentially the same conditions and submitted for inspection at the same time.

4.3. Sampling.

4.3.1. Sampling for examination. Unless otherwise specified in thew contract (see 6.1), for sampling for examination, an “accept at zero” (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. For purposes of examination, as specifies in 4.4.2, samples shall be selected from each lot in accordance with MIL-STD-1916 verification level II or ASQ H1331 index value 4.0.

4.3.2. Sampling for acceptance tests. Unless otherwise specified in thew contract (see 6.1), for sampling for acceptance tests, an “accept at zero” (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. For purposes of testing, as specified in 4.4.2, sampling shall be selected from each lot in accordance with MIL-STD-1916 verification level III or ASQ H1331 index value 2.5.

4.3.3. Sampling of test of protective finish. Sampling for test of protective finish shall be as specified in the applicable specification of 3.3.

4.3.4. Sampling for inspection of packaging and packing. Sampling for examination of packaging, packing and marking shall be specified in the contract order (see 5.1).

4.4. Inspection.

4.4.1. Visual and dimensional examination. Each sample shall be visually and dimensionally examined for conformance with 3.2. If the number of rejected samples exceeds the acceptable number, the lot represented by the sample shall be rejected.

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4.4.2. Acceptance testing. Any sample that fails to conform to the acceptance limits as specified in section 3 herein shall be cause for rejection of the lot represented.

4.5. Tests.

4.5.1. Interlinking. The samples shall be subdivided into groups of two each. An attempt shall be made with each group to manipulate them so that interlocking or interlinking occurs. The washers shall not be bent nor deformed in preforming this test. Any interlinking or interlocking of these washers shall be considered a defect.

4.5.2. Free height test. The free height of the washer shall be measured to the nearest 0.001 inch (see 3.2.4). The washer shall be subjected to a compression, twice pressed flat, released and the free height again measured. The free height after the second compression shall be at least two-thirds of the original free height (or five-sixths) when specified in the invitation for bid.

4.5.3. Twist test. The twist test shall be conducted in accordance with ASME B18.21.1.

4.5.4. Pressure test. The washers shall be clamped between hardened flat steel plates. They shall be held flat for one minute and then released and inspected. The free height of each washer shall be at least two-thirds (or five-sixths, if so specified in the invitation for bids) of the original free height and shall be unimpaired for the lot to be accepted. If all are unimpaired, they shall again be clamped flat, held for 16 hours (or longer), and then released and inspected for the third time. Failure of any sample washer to withstand any of these tests shall be cause for rejection of the lot which it represents.

4.5.5. Hardness test. All sample washers shall be tested for hardness in accordance with the requirements of 3.1. Before making the hardness test on class A washers, both surfaces shall be ground parallel, and decarburization shall be removed.

4.5.6. Decarburization of class A washers. The affected depth of decarburization of carbon steel washers, as specified in 3.1, shall be determined by microscopic examination at 100 magnifications.

4.5.7. Chemical analysis. Samples selected in accordance with 4.3 shall be analyzed to determine compliance with 3.1.1 through 3.1.6, inclusive, as applicable. Failure of any sample to conform to the requirements of this specification shall be cause for rejection of the entire lot represented.

4.5.8. Examination and test of packaging and packing. Examination and test of packaging, packing and marking shall be specified in the contract order (see 5.1).

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5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1. Intended use. The spring lockwashers covered by this specification are intended for automotive and general bolt, nut, and screw applications. They have dual function of (1) spring take-up devices to compensate for developed looseness and the loss of tension between component parts of an assembly, and (2) hardened thrust bearings to facilitate assembly and disassembly of fastening by decreasing the frictional resistance between the bolted surface and the bearing face of the bolt or screwhead or nut.

6.2. Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Class, style of washer required (see 1.2).
c. Type of protective coating, if required (see 3.3.1 through 3.3.7).
d. Military standard part number if applicable (see 3.2.2).
e. Tolerances (see3.2.3).
f. Dimensions (see 3.2.2).
g. Material (see 3.1.1 through 3.1.6).
h. Selection of applicable degrees of preservation and packing required (see 5.1)

6.3. Supersession data. Class A and B style 5- Washer and machine screw assemblies have been deleted from this document. It is suggested that ASME B18.13 for Screw and Washer Assemblies, Sems be used for procurement.

6.4. Subject term (key word) listing.

Cadmium plating

6.5. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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CONCLUDING MATERIAL

Custodians:
Army – AR
Air Force – 99
DLA – GS

Preparing Activity:
DLA – GS7

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